

Q1. Match the List I with List II.

List I (Hydride)		List II (Type of Hydride)
A. NaH	I.	Electron precise
B. PH ₃	II.	Saline
C. GeH ₄	III.	Metallic
D. LaH _{2.87}	IV.	Electron rich

Choose the correct answer from option given below: **(2023)**

- (a) A-III, B-IV, C-II, D-I
- (b) A-II, B-III, C-IV, D-I
- (c) A-I, B-III, C-II, D-IV
- (d) A-II, B-IV, C-I, D-III

Q2. Match List I with List II:

List I		List II
A. Coke	I.	Carbon atoms are sp ³ hybridised
B. Diamond	II.	Used as a dry lubricant
C. Fullerene	III.	Used as a reducing agent
D. Graphite	IV.	Cage like molecules

Choose the correct answer from the options given below: **(2023)**

- (a) A-IV, B-I, C-II, D-III
- (b) A-III, B-I, C-IV, D-II
- (c) A-III, B-IV, C-I, D-II
- (d) A-II, B-IV, C-I, D-III

Q3. The element expected to form largest ion to achieve the nearest noble gas configuration is **(2023)**

- (a) F

- (b) N
- (c) Na
- (d) O

Q4. Match List I with List II:

List I (Oxoacids or Sulphur)	List II (Bonds)
A. Peroxodisulphuric acid	I. Two S-OH, Four S=O, One S-O-S
B. Sulphuric acid	II. Two S-OH, One S=O
C. Pyrosulphuric acid	III. Two S-OH, Four S=O, One S-O-O-S
D. Sulphurous acid	IV. Two S-OH, Two S=O

Choose the correct answer from the option given below: **(2023)**

- (a) A-III, B-IV, C-I, D-II
- (b) A-I, B-III, C-IV, D-II
- (c) A-III, B-IV, C-II, D-I
- (d) A-I, B-III, C-II, D-IV

Q5. Match List I with List II:

List I (Compounds)	List II (Molecular formula)
A. Borax	I. NaBO ₂
B. Kernite	II. Na ₂ B ₄ O ₇ ·4H ₂ O
C. Orthoboric acid	III. H ₃ BO ₃
D. Borax bead	IV. Na ₂ B ₄ O ₇ ·10H ₂ O

Choose the correct answer from the options given below: **(2022)**



- (a) A-I, B-III, C-IV, D-II
- (b) A-IV, B-II, C-III, D-I
- (c) A-II, B-IV, C-III, D-I
- (d) A-III, B-I, C-IV, D-II

Q6. Which of the following reactions is a part of the large-scale industrial preparation of nitric acid? **(2022)**

- (a) $\text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O} \xrightarrow{\text{Pt}, 500\text{K}, 9\text{bar}} 4\text{HNO}_3 + \text{Cu}$
- (b) $\text{NaNO}_3 + \text{H}_2\text{SO}_4 \xrightarrow{\text{Pt}, 500\text{K}, 9\text{bar}} \text{NaHSO}_4 + \text{HNO}_3$
- (c) $4\text{NH}_3 + 5\text{O}_2 \text{ (from air)} \xrightarrow{\text{Pt}, 500\text{K}, 9\text{bar}} 4\text{NO} + 6\text{H}_2\text{O}$
- (d) $4\text{HPO}_3 + 2\text{N}_2\text{O}_5 \xrightarrow{\text{Pt}, 500\text{K}, 9\text{bar}} 4\text{HNO}_3 + \text{P}_4\text{O}_{10}$

Q7. $\text{Na}_2\text{B}_4\text{O}_7 \xrightarrow{\text{heat}} \text{X} + \text{NaBO}_2$

In the above reaction the product "X" is: **(2022)**

- (a) NaB_3O_5
- (b) H_3BO_3
- (c) B_2O_3
- (d) $\text{Na}_2\text{B}_2\text{O}_5$

Q8. Match List I with List II.

List I
(Hydrides)

List II
(Nature)

A. MgH_2

I. Electro
n

B. GeH_4

II. Electro
n
deficien
t

C. B_2H_6

III. Electro
n rich

D. HF

IV. Ionic

Choose the correct answer from the options given below: **(2022)**

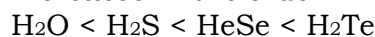
- (a) A-IV, B-I, C-II, D-III
- (b) A-III, B-I, C-II, D-IV
- (c) A-I, B-II, C-IV, D-III
- (d) A-II, B-III, C-I, D-IV

Q9. Which of the following statement is not correct about diborane? **(2022)**

- (a) There are two 3-centre-2-electron bonds
- (b) The four terminal B-H bonds are two centre two electron bonds
- (c) The four terminal Hydrogen atoms and the two Boron atoms lie in one plane
- (d) Both the Boron atoms are sp^2 hybridised

Q10. Given below are two statements.

Statement I: The boiling points of the following hydrides of group 16 elements increases in the order-



Statement II: The boiling points of these hydrides increase with increase in molar mass.

In the light of above statements, choose the most appropriate answer from options given below: **(2022)**

- (a) Both Statement I and Statement II are correct.
- (b) Both Statement I and Statement II are incorrect.
- (c) Statement I is correct but Statement II is incorrect.
- (d) Statement I is incorrect but Statement II is correct.

Q11. Identify the correct statements from the following: **(2020)**

- (1) $\text{CO}_2(\text{g})$ is used as refrigerant for ice-cream and frozen food.
 - (2) The structure of C_{60} contains twelve six carbon rings and twenty five carbon rings.
 - (3) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
 - (4) CO is colorless and odourless gas.
- (a) 1 and 3 only
 - (b) 2 and 3 only
 - (c) 3 and 4 only
 - (d) 1, 2 and 3 only

Q12. Match the following : **(2020)**

Which of the following is correct option?

- (A) (B) (C) (D)
- (a) (ii) (i) (iv) (iii)
- (b) (iii) (iv) (i) (ii)
- (c) (iv) (iii) (ii) (i)
- (d) (i) (ii) (iii) (iv)

Oxide		Nature	
A.	CO	(i)	Basic
B.	BaO	(ii)	Neutr al
C.	Al_2O_3	(iii)	Acidic
D.	Cl_2O_7	(iv)	Amph oteric

Q13. Which of the following is not correct about carbon monoxide? **(2020)**

- (a) It reduces oxygen carrying ability of blood.
- (b) The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.
- (c) It is produced due to incomplete combustion.
- (d) It forms carboxyhaemoglobin.

Q14. Which of the following oxide is amphoteric in nature? **(2020 Covid Re-NEET)**

- (a) SiO_2
- (b) GeO_2
- (c) CO_2
- (d) SnO_2



Q15. Which of the following species is not stable? (2019)

- (a) $[SiF_6]^{2-}$
- (b) $[GeCl_6]^{2-}$
- (c) $[Sn(OH)_6]^{2-}$
- (d) $[SiCl_6]^{2-}$

Q16. Which of the following is incorrect statement? (2019)

- (a) PbF_4 is covalent in nature
- (b) $SiCl_4$ is easily hydrolysed
- (c) GeX_4 ($X = F, Cl, Br, I$) is more stable than GeX_2
- (d) SnF_4 is ionic in nature

Q17. The correct order of atomic radii in group 13 elements is (2018)

- (a) $B < Al < In < Ga < Tl$
- (b) $B < Al < Ga < In < Tl$
- (c) $B < Ga < Al < In < Tl$
- (d) $B < Ga < Al < Tl < In$

Q18. Which one of the following elements is unable to form MF_6^{3-} ion? (2018)

- (a) Ga
- (b) Al
- (c) In
- (d) B

Q19. It is because of inability of ns^2 electrons of the valence shell to participate in bonding that : (2017-Delhi)

- (a) Sn^{4+} is reducing while Pb^{4+} is oxidising
- (b) Sn^{2+} is reducing while Pb^{4+} is oxidising
- (c) Sn^{2+} is oxidising while Pb^{4+} is reducing
- (d) Sn^{2+} and Pb^{2+} are both oxidising and reducing

Q20. The tendency to form monovalent compounds among the Group 13 elements is correctly exhibited in: (2017-Gujarat)

- (a) $B \approx Al \approx Ga \approx In \approx Tl$
- (b) $B < Al < Ga < In < Tl$
- (c) $Tl < In < Ga < Al < B$
- (d) $Tl \approx In < Ga < Al < B$

Q21. Boric acid is an acid because its molecule: (2016-II)

- (a) Accepts OH^- from water releasing proton
- (b) Combines with proton from water molecule
- (c) Contains replaceable H^+ ion
- (d) Gives up a proton



Q22. AlF_3 is soluble in HF only in presence of KF. It is due to the formation of: **(2016-II)**

- (a) $\text{K}_3[\text{AlF}_3\text{H}_3]$
- (b) $\text{K}_3[\text{AlF}_6]$
- (c) AlH_3
- (d) $\text{K}[\text{AlF}_3\text{H}_3]$

Q23. The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence: **(2015 Re)**

- (a) $\text{In} < \text{Tl} < \text{Ga} < \text{Al}$
- (b) $\text{Ga} < \text{In} < \text{Al} < \text{Tl}$
- (c) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
- (d) $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$

Q24. Which of these is not a monomer for a high molecular mass silicone polymer? **(2013)**

- (a) MeSiCl_3
- (b) Me_2SiCl_2
- (c) Me_3SiCl
- (d) PhSiCl_3

Q25. The basic structural unit of silicates is: **(2013)**

- (a) SiO
- (b) SiO_{44-}
- (c) SiO_{32-}
- (d) SiO_{42-}

Q26. Which of the following structure is similar to graphite? **(2013)**

- (a) BN
- (b) B
- (c) B_4C
- (d) B_2H_6



ANSWER KEY

1. Ans. (d)
2. Ans. (b)
3. Ans. (b)
4. Ans. (a)
5. Ans. (b)
6. Ans. (c)
7. Ans. (c)
8. Ans. (a)
9. Ans. (d)
10. Ans. (b)
11. Ans. (c)
12. Ans. (a)
13. Ans. (b)
14. Ans. (d)
15. Ans. (d)
16. Ans. (a)
17. Ans. (c)
18. Ans. (d)
19. Ans. (b)
20. Ans. (b)
21. Ans. (a)
22. Ans. (b)
23. Ans. (c)
24. Ans. (c)
25. Ans. (b)
26. Ans. (a)



SOLUTION

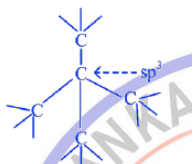
1. Ans.(d)

$\text{LaH}_{2.87} \rightarrow$ Non-stoichiometric
 $\rightarrow$ Metallic/Interstitial hydride.

2. Ans.(b)

Coke: It is used as reducing agent in carbon reduction methods. (in metallurgical process)

Diamond: It is a allotrope of carbon in which each carbon is sp^3 hybridised.



Fullerene: It contains pentagonal & hexagonal rings (cage like structure)

Graphite: It is soft solid because graphite layers are bonded with weak Vander Wall attractions.

3. Ans.(b)

F^{-1} , N^{-3} , Na^{+} and O^{-2}

all ions are isoelectronic containing $10e^{-}$

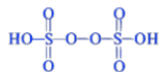
$Z_{\text{eff}} \rightarrow \text{Na}^{+} > \text{F}^{-} > \text{O}^{-2} > \text{N}^{-3}$

order of radius $\rightarrow \text{N}^{-3} > \text{O}^{-2} > \text{F}^{-} > \text{Na}^{+}$

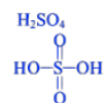
$\rightarrow$ Nitrogen to achieve Noble gas configuration is $\text{tan } 3e^{-}$ and form N^{-3}

4. Ans.(a)

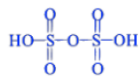
A $\rightarrow$ Peroxodisulphuric acid



B $\rightarrow$ Sulphuric acid



C $\rightarrow$ Pyrosulphuric acid $\text{H}_2\text{S}_2\text{O}_7$



D $\rightarrow$ Sulphurous acid H_2SO_3



5. Ans.(b)

Borax:

$\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} = \text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$

Kernite: $\text{Na}_2\text{B}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$

Orthoboric acid: $\text{H}_3\text{BO}_3 = \text{B}(\text{OH})_3$

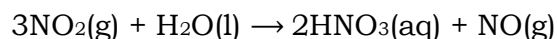
Borax bead: NaBO_2 (Sodium metaborate)

6. Ans.(c)

$4\text{NH}_3 + 5\text{O}_2$ (from air) $\xrightarrow{\text{Pt}, 500\text{K}, 9\text{bar}}$ $4\text{NO} + 6\text{H}_2\text{O}$

$2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$





This is industrial method of preparation of nitric acid.

7. Ans.(c)



Product X is B_2O_3

8. Ans.(a)

A-IV, B-I, C-II, D-III

9. Ans.(d)

Each boron atoms in diborane uses sp^3 hybrid orbitals for bonding.

10. Ans.(b) Boiling point

Compound (K)

H_2O 373

H_2S 213

H_2Se 232

H_2Te 269

The boiling points of these hydrides not exactly increases with increase in molar mass. H_2O has maximum boiling point due to intermolecular hydrogen bonding.

11. Ans.(c)

Dry ice, $\text{CO}_2(\text{s})$, is used as refrigerant

C_{60} contains 20 six membered rings, 12 five membered rings statement 3 and 4 are correct.

12. Ans.(a)

CO is Neutral oxide

BaO is Basic oxide

Al_2O_3 is Amphoteric oxide

Cl_2O_7 is Acidic oxide

13. Ans.(b)

Carbon monoxide binds with Hb to form 300 times more stable compound carboxyhaemoglobin than oxyhaemoglobin complex.

14. Ans.(d)

CO_2 : acidic

SnO_2 : amphoteric

SiO_2 : acidic

GeO_2 : acidic

15. Ans.(d)

Due to presence of vacant d-orbital in Si, Ge and Sn they form species like SiF_6^{2-} , $[\text{GeCl}_6]^{2-}$, $[\text{Sn}(\text{OH})_6]^{2-}$.

SiCl_6^{2-} does not exist because six large chloride ions cannot be accommodated around Si^{4+} due to its small size.



16. Ans.(a)

PbF_4 is ionic in nature, because cation is bigger and anion is smaller.

SiF_4 is easily hydrolysed because it has unoccupied 3d orbital that are able to accept electron pairs from the oxygen atoms on water to form bond. $\text{SiCl}_4 + 3\text{H}_2\text{O} \rightarrow \text{H}_2\text{SiO}_3 + 4\text{HCl}$

GeX_4 is more stable than GeX_2 as in GeX_4 all the orbitals are fully filled.

SnF_4 is ionic in nature as F atom is very small and Sn atom is very large. So, it is ionic in nature according to Fajan's rule.

17. Ans.(c)

Group 13 order of atomic radius is not regular due to transition contraction.

So order is: $\text{B} < \text{Ga} < \text{Al} < \text{In} < \text{Tl}$

18. Ans.(d)

$\therefore$ 'B' has no vacant d-orbitals in its valence shell, so it can't extend its covalency beyond 4, i.e., 'B' cannot form the ion like MF_{63-} i.e. BF_{63-} .

Hence, the correct option is (d).

19. Ans.(b)

Sn^{2+} is reducing while Pb^{4+} is oxidizing. This is because of absence of f-orbital (fully filled) in Sn^{2+} and presence in Pb^{4+} due to which Pb^{4+} show inert pair effect but not Sn^{2+} .

20. Ans.(b)

+1 oxidation state increases down the group as the fully filled d- and f- orbitals will get added due to which inert pair effect will come into role.

$\text{Tl} > \text{In} > \text{Ga} > \text{Al} > \text{B}$

21. Ans.(a)

Boric acid is a Lewis acid, it accepts a pair of electron in aqueous solution to complete its octet because it is an electron deficient compound. $\text{B}(\text{OH})_3 + \text{H}_2\text{O} \rightleftharpoons [\text{B}(\text{OH})_4]^- + \text{H}^+$

22. Ans.(b)

$\text{AlF}_3 + 3\text{KF} \cdot \text{HF} \rightarrow \text{K}_3[\text{AlF}_6]$

AlF_3 is insoluble in Anhydrous HF because F^- ion are not available in Hydrogen bonded HF.

23. Ans.(c)

Order of oxidation state in group 13 is $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$

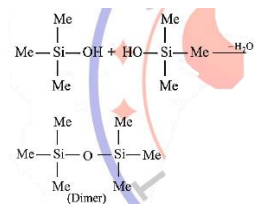
(+1) oxidation state stability increases down the group because of inert pair effect.

24. Ans.(c)

Me_3SiCl is not a monomer for high molecular mass silicone polymer because it generates Me_3SiOH when subjected to hydrolysis which contains only one reacting site.

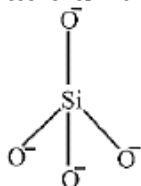
Hence, the polymerization stops just after first step.





25. Ans.(b)

Basic unit for silicates is SiO_4^{4-}



26. Ans.(a)

BN is also known as inorganic graphite with sp^2 hybridisation and hexagonal structure.

